EXECUTIVE & EDUCATIONAL BRIEF • SPOKEN TUTORIAL, IIT BOMBAY

Spoken Social — Product & Application Brief

Empowering Open-Source Learning Through Gamification, Real-Time Visibility, and Peer Community

Platform Overview at a Glance

Mission: Transform solitary software learning into an engaging, trackable, and social journey.

Audience: College students, faculty coordinators, and institutional training heads across India.

Live Scale: Currently active in production, serving 5,400+ students across 170+ institutions.

Ecosystem: Companion platform directly integrated with Spoken Tutorial (spoken-tutorial.org), IIT Bombay.

1. Executive Summary & Why Spoken Social Exists

Spoken Tutorial (IIT Bombay) has revolutionized self-paced software training across India, distributing high-quality audio-video tutorials in regional languages for tools like Python, Linux, C++, Scilab, and LibreOffice. However, self-paced distance learning faces a universal challenge: silent drop-off.

When students encounter obstacles—such as software installation failures, complex syntax errors, or lack of computer lab access—they often lose momentum in isolation and stop practicing. Because conventional learning platforms only record initial course enrollment and final exam certificates, instructors and coordinators have zero visibility into who has stalled until months later when batches finish.

Spoken Social solves this by serving as 'Duolingo meets LinkedIn meets a Collaborative Learning Community'. It tracks every step of the learning journey, rewards effort with points and levels, provides faculty with real-time early warning dashboards, and fosters peer collaboration.

2. Who the Platform Serves

User Persona	Core Needs & Pain Points	Spoken Social Solution
The Student (e.g., Priya, 2nd-Yr B.Tech)	Needs engagement, clear daily targets, friendly competition, instant doubt help, and a portfolio to prove software skills to recruiters.	Gamified dashboard, step-by-step progress checklists, XP and levels, campus leaderboards, and forum rewards.
The Faculty Coordinator (e.g., Prof. Sharma, SPOC)	Coordinates training for 200+ students across departments; needs to know who is actively practicing and who needs help before exams.	Live institution radar, student drill-down, friction analysis (installation bugs), and 1-click Excel compliance reports.
The Programme Admin (IIT Bombay / EduPyramids)	Needs oversight across dozens of universities, multiple concurrent training batches, and national course completion trends.	Platform-wide drill-down analytics, cross-institutional benchmarks, and seamless single sign-on integration.

3. Core Features Live Today

A. For Students: The Gamified Learning Journey

- **Interactive Step-by-Step Tracker:** Breaks every course into bite-sized steps: Watched the tutorial, Practiced hands-on, Completed assignment, and Attempted mini-task. Progressive unlocking ensures students do not skip crucial practice.
- **Transparent Points (XP) & Levels:** Earn points for genuine learning actions: +10 pts for watching, +10 pts for software practice, +5 pts for assignments, +5 pts for asking doubts, and +10 pts for answering classmates. Learners advance through 6 tiers (from Beginner to Spoken Legend).
- **Dynamic Leaderboards:** Clean, friendly rankings across 3 scopes (Your College, Your Course, and National) across Weekly, Monthly, and All-Time windows, with top-3 podium highlights.
- **Smart Roadblock Detection:** When a student indicates 'No' on practicing or an assignment, a non-intrusive prompt captures the exact roadblock (software installation error, lab unavailable, difficult concept, time constraint). This transforms a failure into a teacher support alert.
- **Learner Profile:** Displays earned XP, rank badges, completed courses, and learning streaks.

B. For Faculty Coordinators: Real-Time Command Center

- **Live Cohort Radar:** Instantly categorizes enrolled students into four clear cohorts: Completed (100% finished), Active (steady recent progress), At-Risk (slowing down/behind schedule), and Stalled (no activity recorded).
- **Individual Student Drill-Down:** Coordinators can click on any student to view their exact timeline and identify precisely which lesson or command stopped them.
- **Friction Analysis Heatmaps:** Aggregates student feedback to highlight class-wide roadblocks (e.g., 40% of the class cannot install Linux dependencies), allowing faculty to schedule targeted lab sessions.
- **One-Click Excel Reports:** Download fully formatted progress sheets for department meetings, accreditation bodies (NAAC/NBA), and Semester Training Planner (STP) audits in one click.

C. Seamless Ecosystem Integration

- **Single Sign-On (SSO):** Students sign in with their existing Spoken Tutorial credentials. No new passwords, accounts, or friction.
- **Discussion Forum Sync:** Tied directly into the Spoken Tutorial forums to encourage peer support with built-in spam prevention.

4. Learning Activities Hosted on the Platform

Learning Activity	Educational Objective	Platform Mechanism
1. Video Screencast Learning	Conceptual Comprehension	Students watch self-paced video lessons with voice narration in regional languages or English across programming, Linux, Scilab, and CAD tools. Earns +10 XP.
2. Guided Lab Practice	Hands-on Application	The heart of Spoken Tutorial: side-by-side software execution where students replicate commands and code on their own PC or college lab. Earns +10 XP.
3. Formative Assignments	Knowledge Application	Problem-solving tasks at the end of each tutorial (e.g., writing a script, configuring file permissions, generating computational graphs). Earns +5 XP.

Learning Activity	Educational Objective	Platform Mechanism
4. Practical Mini-Tasks	Advanced Challenges	Optional creative challenge tasks for students who want to explore beyond basic curriculum requirements.
5. Peer Q&A & Mentorship	Collaborative Learning	Asking technical doubts (+5 XP) and answering classmates (+10 XP) on Spoken Tutorial forums. Reinforces knowledge through teaching.
6. Diagnostic Roadblock Logs	Reflective Feedback	Students self-report hurdles when stuck, turning moments of friction into actionable support tickets for faculty.
7. Structured Cohort Workshops	Institutional Training	Time-bound batches (e.g., 'Semester 3 Python Workshop' or 'Faculty FDP on Scilab') with organized schedules, deadlines, and mentor oversight.
8. Certification Exam Prep	Summative Assessment	Readiness milestones and practice checks leading into official proctored Spoken Tutorial online certification exams.

5. What More Can Be Added (Future Roadmap & Expansion)

Spoken Social has been architected to expand seamlessly. Below are key high-impact feature enhancements ready for upcoming phases:

A. Ready in Codebase (Immediate Activation)

- **Campus Social Feed:** A clean, community wall where students can share mini-projects, code snippets, study resources, and celebrate learning wins with likes and comments.

B. Gamification & Habit-Building Enhancements

- **Learning Streaks & Daily Habits:** Daily study streaks ('🔥 15-day streak!') with 'Streak Freezes' to keep students returning daily and build enduring learning habits.
- **Achievement Badges:** Collectible visual achievement badges for key milestones (e.g., 'First Script Executed', 'Weekend Warrior', 'Helpful Mentor', 'Top 5 Campus Performer').

C. Career & Portfolio Features (LinkedIn-Inspired)

- **Digital Learning Resume:** Public, shareable student profile showcasing completed courses, verified certificates, code repositories, and faculty endorsements—acting as a modern tech resume.
- **Direct Job & Internship Pathways:** Connecting high-scoring students directly with internship and junior developer openings via Spoken Tutorial's Job Recommendation System (JRS).

D. AI-Assisted Learning & Smart Support

- **AI Learning Companion:** A 24/7 in-app tutor to help students troubleshoot local software installation errors, explain compiler bug outputs, and answer concept queries in their mother tongue.
- **Automated Dropout Early Warning:** Predictive analytics alerting coordinators automatically when a student shows drop-off patterns, suggesting timely interventions.

E. Collaborative & Mobile Learning

- **Mobile Companion App:** Lightweight mobile app with push notifications for batch deadlines, streak reminders, and weekly leaderboard podium alerts.
- **Cross-College Study Circles:** Virtual study circles allowing students from different colleges enrolled in the same course to collaborate and peer-review code.